

10.2 Exercise Rate Optimisation:

11 Multi-Level Monte Carlo (MLMC)

11.1 Introduction:

In this section, we give an overview of the key results on Multi-Level Monte Carlo from [7] and [8]. When do we use Monte Carlo methods?

1. Uncertainty is of high dimension - which can also mean that the number of uncertain input parameters is high.
2. Uncertainty is strongly non-linear.

11.1.1 Standard (simple) Monte Carlo:

Under standard Monte Carlo the estimator of $\mathbb{E}[P]$ is just an average of values of $P(\omega)$ for N independent samples from the probability space $(\Omega, \mathcal{F}, P)$ given by:

$$\mathbb{E}[\hat{P}] = N^{-1} \sum_{i=1}^N P(\omega^{(n)}) \quad (113)$$

The variance of the estimate is given by $N^{-1}\mathbb{V}[P]$, giving an rms error of $\mathcal{O}(N^{-\frac{1}{2}})$. An accuracy of ϵ therefore needs $N = \mathcal{O}(\epsilon^2)$ samples.

11.1.2 Control Variates:

11.1.3 General Idea for Multi-Level Monte Carlo:

Suppose we have a sequence $P_0, \dots, P_{L-1}$ which approximates P_L with increasing accuracy and cost. Then we can write the expected value of P_L using a telescoping sum:

$$\mathbb{E}[P_L] = \mathbb{E}[P_0] + \sum_{l=1}^L \mathbb{E}[P_l - P_{l-1}] \quad (114)$$

This means that we can use the following unbiased estimator of P_L :

$$Y = N_0^{-1} \sum_{n=1}^{N_0} P_0^{(0,n)} + \sum_{\ell=1}^L \left\{ N_\ell^{-1} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}) \right\}. \quad (115)$$

The total cost and the total variance are then given by (assuming independent samples):

$$\text{Cost} = \sum_{\ell=0}^L N_\ell C_\ell, \quad \mathbb{V}(Y) = \sum_{\ell=0}^L N_\ell^{-1} V_\ell.$$

Assuming a fixed variance, the cost is minimised by choosing N_ℓ to minimise:

$$\sum_{\ell=0}^L (N_\ell C_\ell + \mu^2 N_\ell^{-1} V_\ell),$$

with μ^2 a Lagrange multiplier. This gives:

$$N_\ell = \mu \sqrt{\frac{V_\ell}{C_\ell}},$$

Assuming we want an overall variance given by ϵ^2 we require:

$$\mu = \epsilon^{-2} \sum_{\ell=0}^L \sqrt{V_\ell C_\ell},$$

Hence, the optimal total computational cost is given by:

$$C_{\text{total}} = \epsilon^{-2} \left(\sum_{\ell=0}^L \sqrt{V_\ell C_\ell} \right)^2. \quad (1.1)$$

What is determined by the particular problem is the relative rate at which the variance decreases with level compared to the increase in the computational cost.

11.1.4 Geometric MLMC:

When the sample space is infinite, such as when we want to simulate SDEs, the output on the level P_ℓ on level ℓ is an approximation of the random variable P , which cannot be simulated.

Assuming that Y is an approximation to $\mathbb{E}[P]$, then the Mean Square Error (MSE) is given by:

$$\text{MSE} \equiv \mathbb{E}[(Y - \mathbb{E}[P])^2] = \mathbb{V}[Y] + (\mathbb{E}[Y] - \mathbb{E}[P])^2.$$

where the first contribution to the error is the statistical sampling error which measures the variance from using a finite number of Monte Carlo samples and the second is the weak error (or bias), which arises from stopping the telescoping sum at level L . We then assume that Y is the multi-level estimator given by:

$$Y = \sum_{\ell=0}^L Y_\ell, \quad Y_\ell = \frac{1}{N_\ell} \sum_{n=1}^{N_\ell} (P_\ell^{(\ell,n)} - P_{\ell-1}^{(\ell,n)}), \quad P_{-1} \equiv 0.$$

then we obtain that:

$$\mathbb{E}[Y] = \mathbb{E}[P_L], \quad \mathbb{V}[Y] = \sum_{\ell=0}^L \frac{V_\ell}{N_\ell}, \quad V_\ell \equiv \mathbb{V}[P_\ell - P_{\ell-1}].$$

To ensure that the MSE is less than ϵ^2 , it is sufficient to ensure that each contribution to the MSE is less than ϵ^2 . [7] then assumes a geometric sequence of levels with the cost increasing exponentially with levels and variance decreasing exponentially with levels.

Giles' MLMC Theorem

Theorem 7. *Let P be a random variable and P_ℓ its level- ℓ approximation. Suppose there exist independent estimators Y_ℓ based on N_ℓ samples, each of cost C_ℓ and variance V_ℓ , and positive constants $\alpha, \beta, \gamma, c_1, c_2, c_3$ with $\alpha \geq \frac{1}{2} \min(\beta, \gamma)$ such that*

$$\textbf{Weak Error: } |\mathbb{E}[P_\ell - P]| \leq c_1 2^{-\alpha \ell},$$

$$\mathbb{E}[Y_\ell] = \begin{cases} \mathbb{E}[P_0], & \ell = 0, \\ \mathbb{E}[P_\ell - P_{\ell-1}], & \ell > 0, \end{cases}$$

$$\textbf{Variance: } V_\ell \leq c_2 2^{-\beta \ell},$$

$$\textbf{Cost: } C_\ell \leq c_3 2^{\gamma \ell}.$$

Then for any $\varepsilon < e^{-1}$ there exist L and $\{N_\ell\}$ such that the estimator

$$Y = \sum_{\ell=0}^L Y_\ell$$

satisfies

$$\mathbb{E}[(Y - \mathbb{E}[P])^2] < \varepsilon^2,$$

and its cost C obeys

$$\mathbb{E}[C] \leq \begin{cases} c_4 \varepsilon^{-2}, & \beta > \gamma, \\ c_4 \varepsilon^{-2} (\log \varepsilon)^2, & \beta = \gamma, \\ c_4 \varepsilon^{-2 - (\gamma - \beta)/\alpha}, & \beta < \gamma. \end{cases}$$

Sketch of the proof:

Each of the constraints on the size of the weak error and the statistical error gives a constraint on the number of levels L and the number of samples per level N_ℓ respectively.

1. $\mathbb{E}[Y - P] = \mathbb{E}[\sum_\ell Y_\ell - P] = \mathbb{E}[P_L - P] \leq c_1 2^{-2\alpha L} < \frac{1}{\sqrt{2}}\varepsilon$. This gives that $L \gtrsim \frac{1}{\alpha} \log_2(\frac{1}{\varepsilon})$.
2. We obtain that the optimal N_ℓ is given by: $N_\ell = \frac{2}{\varepsilon^2} (\sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}}) \sqrt{\frac{V_\ell}{C_\ell}}$.

The optimal computational cost is then, assuming that $N_\ell \propto 2^{-(\beta+\gamma)\frac{\ell}{2}}$ and $N_\ell C_\ell \propto 2^{-(\beta+\gamma)\frac{\ell}{2}}$:

$$C \lesssim \frac{1}{\varepsilon^2} \left(\sum_{\ell'=0}^L \sqrt{V_{\ell'} C_{\ell'}} \right)^2 \quad (116)$$

We now have three cases depending on the relative magnitudes of β and γ :

1. $\beta > \gamma$: then we have a sequence that converges so the brackets are constant. Hence $C \lesssim c_4 \varepsilon^{-2}$. In this case, most of the computational cost is from the coarse levels where ℓ is low. To further improve the computational cost we can note that adding the further levels will not significantly add to the computational cost. Hence, the equal error split is sub-optimal. Instead, we may give more error to the statistical error/variance term and so fewer samples N_ℓ will be required.
2. $\beta = \gamma$: then we have $C \lesssim k \varepsilon^{-2} (L+1)^2 \approx c_4 \varepsilon^{-2} \log(\varepsilon)^2$. In this case, computational cost and contribution to the variance is evenly spread across all the levels.
3. $\beta < \gamma$: then the dominant contribution to the sum is from level L . So we obtain: $C \lesssim k \varepsilon^{-2} 2^{-(\beta-\gamma)L} \approx \varepsilon^{-2 - \frac{(\gamma-\beta+\gamma)}{\alpha}}$. In this case the dominant computational cost is from the finest levels.

We note that the computational cost for standard Monte Carlo is $\mathcal{O}(\varepsilon^{-2 - \frac{\gamma}{\alpha}})$.

11.1.5 Multi-Index Monte Carlo (MIMC):

11.2 Multi-Level Monte Carlo for SDEs:

We consider SDE's of the form:

$$\begin{aligned} dS_t &= rS_t dt + \sigma S_t dW_t \\ S_0 &= s_0 \end{aligned} \quad (117)$$

We then want to compute the expected value of $f(S(T))$, where the function f is assumed for now to be Lipschitz.

A simple Euler-Maruyama discretisation with time step h of the SDE gives:

$$\hat{S}_{n+1} = \hat{S}_n + a(\hat{S}_n, t_n) h + b(\hat{S}_n, t_n) \Delta W_n.$$

11.2.1 Standard Monte Carlo:

The standard Monte Carlo estimator for $\mathbb{E}[f(S(T))]$ is given by:

$$\hat{Y} = \frac{1}{N} \sum_{i=1}^N f(\hat{S}_T^{(i)}).$$

where we implicitly assume a certain step-size h . Assuming that $a(S_t, t)$ and $b(S_t, t)$ satisfy certain conditions (Bally and Talay 1995, Kloeden and Platen 1992, Talay and Tubaro 1990) then the MSE is given by:

$$MSE \approx c_1 N^{-1} + c_2 h^2 \quad (118)$$

The first term corresponds to the statistical sampling error (variance in the estimator $\hat{Y}$) and the second term the weak error due to error induced by the Euler time discretisation which is $\mathcal{O}(h)$. Setting $MSE = \epsilon^2$, we require $N = \mathcal{O}(\epsilon^{-2})$ and $h = \mathcal{O}(\epsilon)$. This gives a total computational cost $C_{MC} = \mathcal{O}(\epsilon^{-3})$

11.2.2 Multi-Level Monte Carlo for Euler-discretised SDE:

Giles' MLMC Method:

1. We take a geometric sequence of levels such that at level ℓ the uniform time-step is taken to be $h_\ell = h_0 M^{-\ell}$ with M a positive integer typically taken to be 2. h_0 may be taken to be the entire time interval T as long as it remains a good enough control variate for the other finer levels.
2. To couple the coarse and the fine levels we use the same underlying Brownian path and sum the Brownian increments for the fine path time steps to obtain the Brownian increments for the coarse time-steps.
3. We can then use that given a Lipschitz payoff function f for which for any S_1, S_2 there exists a constant K such that $|f(S_1) - f(S_2)| \leq K \|S_1 - S_2\|$ and that the strong error of the Euler discretisation is $\mathbb{E}[\|S - \hat{S}\|^2] = \mathcal{O}(h)$ that:

$$\begin{aligned} V_\ell &\equiv \mathbb{V}[P_\ell - P_{\ell-1}] = \mathbb{V}[(P_\ell - P) - (P_{\ell-1} - P)] \\ &\leq 2(\mathbb{V}[P - P_\ell] + \mathbb{V}[P - P_{\ell-1}]) \\ &\leq 2(K^2 \mathbb{E}[\|S - \hat{S}_\ell\|^2] + K'^2 \mathbb{E}[\|S - \hat{S}_{\ell-1}\|^2]) \end{aligned} \quad (119)$$

This gives that $V_\ell = \mathcal{O}(h_\ell)$. Hence, if we take $M = 2$ we have $\alpha = 1, \beta = 1, \gamma = 1$. This means the complexity to reach a rms error of ϵ^2 is, by Theorem [7], $C_{MLMC} = \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$.

This means the MLMC method is near the optimal lower bound of $\mathcal{O}(\epsilon^2)$ for complexity, save logarithmic factors.

11.2.3 Numerical Example: Pricing a European Call Option under GBM

We now price a European Call Option on a single stock, giving pay-off $f(S_T) = e^{-rT}(S_T - K)^+$ whose underlying asset satisfies Geometric Brownian Motion (GBM) following the methodology outlined in [7].

$$dS_t = rS_t + \sigma S_t dW_t \quad (120)$$

Following [7], we choose $r = 0.05, \sigma = 0.2, S_0 = 100, T = 1, K = 100$, allowing us to verify that we do in fact correctly implement MLMC.

This contrasts with the standard Monte Carlo cost which is approximately $\epsilon^{-2} C_L V_0$.

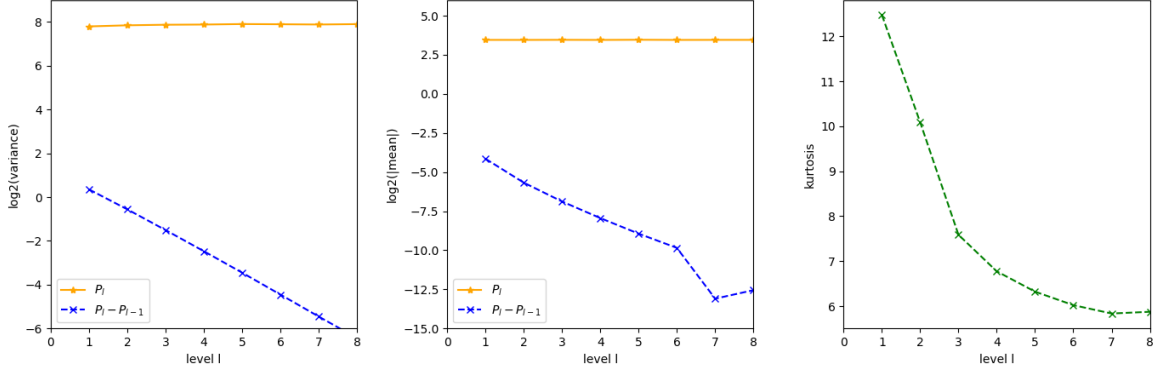


Figure 2: We estimate the variance, absolute mean and kurtosis with level ℓ of an initial sample of 10^5 paths. [7,8] checks the kurtosis given by $\kappa = \frac{\mathbb{E}[X^4]}{(\mathbb{E}[X^2])^2}$ for a random variable X with zero mean. For a large number of sample, the standard deviation of the sample variance of the random variable X is $\sigma_{samplevar} = \sqrt{\frac{\kappa-1}{N} \mathbb{E}[X^2]}$. It is important to check that κ is small to ensure that a reliable estimate of the variance is obtained.

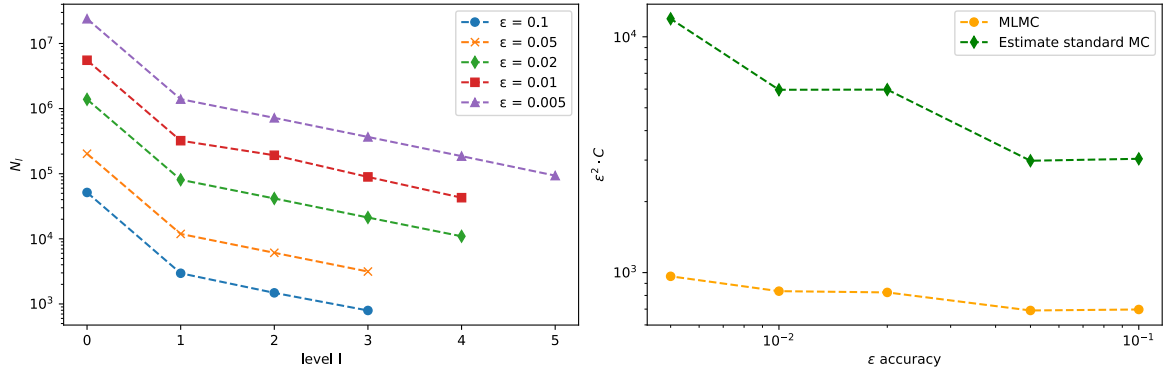


Figure 3: Left hand figure plots number of paths per level for a selected accuracy ϵ . Right hand figure plots $\epsilon^2 * C$ for MLMC and an estimated $\epsilon^2 * C$ of standard Monte Carlo. Given that the theoretical expected cost is for MLMC for our case is $C \sim \mathcal{O}(\epsilon^{-2}(\log(\epsilon))^2)$ we expect that $\epsilon^2 * C$ should be approximately constant as we see in the plot.

11.2.4 Discontinuous Payoffs:

11.3 Multi-Level Weighted Least Squares Approximation [1] [2]

11.3.1 Polynomial (Weighted) Least Squares Approximation:

The goal of polynomial weighted least squares (PWLS) is to approximate a function f , $f \in \mathcal{L}_\mu^2(\Gamma)$, where μ is a probability measure on parameter space Γ , using out finitely many point evaluations of f . The general methodology goes as follows:

1. Choose a finite-dimensional space of polynomials $V \in \mathcal{L}_\mu^2(\Gamma)$.
2. Compute $f(\vec{y}_i)$ for random samples $(\vec{y}_i)_{i=1}^N$ drawn i.i.d from some distribution ν on Γ
3. Minimise the discrete least squares error:

$$\begin{aligned} \Pi_V f &:= \arg \min_{v \in V} \|f - v\|_{disc}^2 \\ \|f - v\|_{disc}^2 &:= \frac{1}{N} \sum_{j=1}^N \frac{d\mu}{d\nu} |f(\vec{y}_j) - v(\vec{y}_j)|^2 := \langle f - v, f - v \rangle_{disc} \end{aligned} \quad (121)$$

where the $\frac{d\mu}{d\nu} := w$ is the weight function and corrects for the difference between the true distribution μ and the sampling distribution ν .

4. The coefficients $\vec{c}$ of $\Pi_V f$ with respect to any basis $(B_j)_{j=1}^m$ of V are given by the normal equation:

$$\hat{G}\vec{c} = \vec{v} \quad (122)$$

with $\hat{G}_{ij} = \langle B_i, B_j \rangle_{disc}$ and $v_j = \langle f, B_j \rangle_{disc}$ with $i, j \in \{1, \dots, m\}$ assuming that $\hat{G}$ is invertible. If $\hat{G}$ is not invertible, then we define $\Pi_V f$ as the one with minimal $\mathcal{L}_\mu^2(\Gamma)$ norm.

Uniqueness of minimisation requires that $N \geq \dim(V) = m$. If $\nu = \mu$, then $N = \mathcal{O}(m^b \log(m))$ ensures a quasi-optimal polynomial fit, where the specific value of $b \geq 1$ depends on μ . For a specific optimal distribution we can obtain the same result with $N = Cm \log(m)$ (Hampton and Doonstan (2015)).

11.3.2 Sampling from the optimal distribution:

We want to sample from the optimal distribution ν_V^* for which $N = \mathcal{O}(m \log(m))$. This occurs for the weight $w_V^* = \rho_V^{*-1}$ which is associated with density:

$$\rho_V^*(y) := \frac{1}{m} \sum_{j=1}^m |B_j(y)|^2.$$

[1] show that the density of the optimal distribution ν_V^* associated with any downward closed² polynomial subspace on $[0, 1]^d$ with respect to the Lebesgue measure $d\lambda$ satisfies:

$$C^{-d} < \frac{d\nu_V^*}{d\lambda} \leq C^d p_d^\infty,$$

with $0 < C < \infty$ and $p_d^\infty(y)$, the Lebesgue density of the d-dimensional arcsine distribution given by:

$$p_d^\infty(y) := \prod_{j=1}^d \frac{1}{\pi \sqrt{y_j(1-y_j)}}.$$

²A subspace is downward closed if you have monomial in your subspace and there is another monomial whose degree is smaller in each component, then this monomial is also in your subspace.

The upper bound implies that the number samples for a near-optimal approximation of f , assuming the weight function associated with the arcsine distribution, is given by $C^d m \log(m)$, where C is independent of V (but depends exponentially on d). C^d has been tested to be small even in high dimensions, (≈ 4) for $d = 10$. **As a consequence the upper bound allows for efficient sampling in many cases (without the need for MCMC methods or acceptance/rejection sampling) from the arcsine distribution rather than the optimal distribution for all polynomial subspaces. It also gives a simple weight function independent of V , which is important for adaptive algorithms.**

One way to sample from the univariate arcsine distribution:

1. Generate a uniformly distributed random variable $X \sim U[-\frac{\pi}{2}, \frac{\pi}{2}]$.
2. Transform X via: $Y = \frac{1}{2}(1 + \sin(X))$ then Y samples the univariate arcsine distribution.

11.3.3 Single-Level Approach to Inexact Solutions:

The single-level approach applies the weighted least squares approximation to a single f_n approximating some function f . The problem with this approach is that a good approximation requires a large subspace V and fine time steps leading to many, expensive samples.

It may be shown that if the following two assumptions hold:

1. There are $\beta, \gamma > 0$ such that $\|f - f_h\|_{\mathcal{L}_\mu^2(\Gamma)} \lesssim h^\beta$ and each evaluation of f_h requires work $h^{-\gamma}, \forall h > 0$.
2. There is an $\alpha > 0$ and a sequence of subspaces $(V_n)_{n=1}^\infty$ such that $\inf_{v \in V_n} \|f_h - v\|_{\mathcal{L}^\infty(\Gamma)} \lesssim (\dim(V_n))^{-\alpha}$ for all $h > 0, n \in \mathbb{N}$

then an accuracy of $\epsilon > 0$ requires $h \approx \epsilon^{-\frac{\gamma}{\beta}}$ and $\dim(V_n) \approx \epsilon^{-\frac{1}{\alpha}}$. The total work therefore is given by:

$$\begin{aligned} &\approx h^{-\gamma} \dim(V_n) \log(\dim(V_n)) \\ &\approx \epsilon^{-\frac{\gamma}{\beta}} \epsilon^{-\frac{1}{\alpha}} |\log(\epsilon)| \end{aligned} \tag{123}$$

Where $\epsilon^{-\frac{\gamma}{\beta}}$ is the contribution to the work from PDE solutions and $\epsilon^{-\frac{1}{\alpha}}$ the contribution from the complexity of polynomial spaces.

11.4 Multi-Level Approach to Inexact Solutions

11.4.1 Core Assumptions:

[1,2] show how the Multi-Level methodology can be applied to improve the computational work required for a weighted least squares approximation.

Authors consider the approximation of a function $f_\infty : \Gamma \subset \mathbb{R}^d \rightarrow \mathbb{R}, d \in \mathbb{N} \cup \{\infty\}$ in a normed vector space $(F, \|\cdot\|_F) \hookrightarrow (L_\mu^2(\Gamma), \|\cdot\|_{L_\mu^2(\Gamma)})$ of continuous functions on Γ , under the following assumptions:

A1 (Convergence of approximations). There exist functions $f_n \in F, n \geq 1$ such that

$$\begin{aligned} \|f_\infty - f_n\|_F &\lesssim n^{-\beta_s} \\ \|f_\infty - f_n\|_{L_\mu^2(\Gamma)} &\lesssim n^{-\beta_w} \end{aligned}$$

for some $\beta_s > 0$ and $\beta_w \geq \beta_s$.

A2(p) (Polynomial approximability). There exist downward closed spaces of polynomials $V_m, m \geq 1$ on Γ such that

$$\dim V_m \lesssim m^\sigma, \quad e_{m,p}(F) \lesssim m^{-\alpha}$$

for some $\sigma > 0$, $\alpha > 0$, and $p = 2$ or $p = \infty$, where

$$e_{m,2}(F) := \sup_{f \in F} \frac{e_{V_m, \|\cdot\|_{L^2_\mu(\Gamma)}}(f)}{\|f\|_F}, \quad e_{m,\infty}(F) := \sup_{f \in F} \frac{e_{V_m, \|\cdot\|_\infty}(f)}{\|f\|_F}.$$

We can understand this assumption as follows: there is a sensible polynomial hierarchy V_m whose size grows like m^σ and for which the best weighted L^p approximation error of any $f \in F$ decays like $m^{-\alpha}$ once normalised by $\|f\|_F$.

A3 (Sample work). The work required for a single evaluation of f_n satisfies

$$\text{Work}(f_n) \lesssim n^\gamma$$

for some $\gamma > 0$.

11.4.2 Multi-Level Estimator:

We let $h_l \rightarrow 0$ (which authors take to be the finite element of diameter l^{-1} but could in principle also be time step of an SDE) in the limit $l \rightarrow \infty$ and consider a sequence of functions $f_{h_0}, \dots, f_{h_l}$ which approximate in some suitable sense a function f . We then write f as a telescoping sum:

$$f = f_{h_0} + \sum_{l=1}^{\infty} (f_{h_l} - f_{h_{l-1}}) \quad (124)$$

As before, we cut-off this sum at some suitable L . Note that the coarsest approximation of f , f_{h_0} , is taken to be of the same order as f and it is cheap to evaluate many samples. Further, $(f_{h_l} - f_{h_{l-1}}) \rightarrow 0$ in the limit $l \rightarrow \infty$ and samples are (increasingly) expensive to evaluate.

Choosing that $(h_l)_{l=1}^\infty$ is exponentially decreasing and our sequence of subspaces $(V_l)_{l=1}^\infty$ to be exponentially increasing we can write:

$$\mathcal{S}_L(f) := \Pi_{V_L} f_0 + \sum_{l=1}^L \Pi_{V_{L-l}} (f_l - f_{l-1}) \quad (125)$$

where the operator $\Pi_{V_l} : F \rightarrow V_l$, for $0 \leq l \leq L$, is the random weighted least square approximation which projects onto the subspace V_l . [Because the projection at each level differs this sum does not telescope in the same way as \(124\), nor is it required to do so \(all that is required is that the polynomial approximability assumption is fulfilled\).](#) This entails we get error contributions from each level l . However, in the asymptotic limit $l \rightarrow \infty$ V_l goes to infinity and $f_l - f_{l-1} \rightarrow 0$. This entails that our error goes to zero exponentially fast.

11.4.3 Multi-Level Convergence Analysis

Multi-Level Convergence Analysis

If $\|f - f_h\|_F \lesssim h^\beta$ and each evaluation of f_h requires work $h^{-\gamma}$, $\forall h > 0$ for some space F for which $\inf_{v \in V_l} \|g - v\|_{L^\infty(\Gamma)} \lesssim \|g\|_F (\dim V_l)^{-\alpha}$, $\forall g \in F$, $l \in \mathbb{N}$, then the multilevel polynomial approximation (with optimal sampling) requires the work

$$W \simeq \varepsilon^{-\max\{\gamma/\beta, 1/\alpha\}} |\log \varepsilon|^r,$$

(with some $r = r(\alpha, \beta, \gamma) > 0$ whose value is known) to achieve

- *Convergence in probability:*

$$\|S_L(f) - f\|_{L^2_\mu(\Gamma)} \leq \varepsilon \quad (126)$$

with probability $> 1 - \varepsilon^{|\log \varepsilon|}$, $0 < \varepsilon \ll 1$.

- *Mean square convergence* (if L^∞ bound is available):

$$\mathbb{E} \|\tilde{S}_L(f) - f\|_{L^2_\mu(\Gamma)}^2 \leq \varepsilon^2.$$

Here we follow [2] in considering the particular case where $\alpha, \gamma, \beta = 1$ and so we aim to show that $W \simeq \varepsilon^{-1} |\log(\varepsilon)|^r$, for some r depending on α, β, γ

Proof. First we derive an error bound. For any $L > 0$ we have

$$\begin{aligned} \|S_L(f) - f_{h_L}\|_{L^2_\mu} &= \left\| \Pi_{V_L} f_0 + \sum_{l=1}^L \Pi_{V_{L-l}} (f_l - f_{l-1}) - \left(f_0 + \sum_{l=1}^L (f_l - f_{l-1}) \right) \right\|_{L^2_\mu} \\ &= \left\| (\Pi_{V_L} - \text{Id}) f_0 + \sum_{l=1}^L (\Pi_{V_{L-l}} - \text{Id}) (f_l - f_{l-1}) \right\|_{L^2_\mu} \\ &\leq \underbrace{\|\Pi_{V_L} - \text{Id}\|_{F \rightarrow L^2}}_{\text{operator norm}} \|f_0\|_F + \sum_{l=1}^L \|\Pi_{V_{L-l}} - \text{Id}\|_{F \rightarrow L^2} \|f_l - f_{l-1}\|_F \\ &\leq \|\Pi_{V_L} - \text{Id}\|_{F \rightarrow L^2} \|f_0\|_F + \sum_{l=1}^L \|\Pi_{V_{L-l}} - \text{Id}\|_{F \rightarrow L^2} \|f_l - f\|_F. \end{aligned}$$

Assuming $\dim V_l = e^l$ and $h_l = e^{-l}$, one has $\|\Pi_{V_l} - \text{Id}\| \approx e^{-l} \approx \|f_l - f\|_F$. Hence

$$\begin{aligned} \|S_L(f) - f\| &\leq \|S_L(f) - f_{h_L}\| + \|f_{h_L} - f\| \\ &\lesssim e^{-L} + \sum_{l=1}^L e^{-(L-l)} e^{-l} + e^{-L} = e^{-L} (L + 2). \end{aligned}$$

We note that in the asymptotic limit $L \rightarrow \infty$ the error goes to 0. Next we derive the computational complexity of the Multi-Level Polynomial:

$$\begin{aligned} \text{Work}(S_L(f)) &= \text{Work}(\Pi_{V_L} f_0) + \sum_{l=1}^L \text{Work}(\Pi_{V_{L-l}} (f_l - f_{l-1})) \\ &\approx \dim V_L \log(\dim V_L) \text{Work}(f_0) + \sum_{l=1}^L \dim V_{L-l} \log(\dim V_{L-l}) \text{Work}(f_l - f_{l-1}) \\ &\lesssim e^L L + \sum_{l=1}^L e^{L-l} (L-l) e^l \\ &\approx e^L (L+1)^2. \end{aligned}$$

Denoting the right-hand side of the error bound $e^{-L}(L+2) = \varepsilon$, we may rephrase the work bound as

$$\text{Work}(S_L(f)) \lesssim \varepsilon^{-1} |\log \varepsilon|^3 = \varepsilon^{-\max\{\gamma/\beta, 1/\alpha\}} |\log \varepsilon|^3.$$

□

11.5 Application of Multi-Level Least Squares Polynomial Regression Method to SDEs:

11.5.1 Original Example: Random PDEs

Let $f(\mathbf{y}) = Q(u_{\mathbf{y}})$, where $u_{\mathbf{y}}$ is the solution to a PDE with random parameters $\mathbf{y}$. For example:

$$\begin{aligned} -\nabla \cdot (a(x, \mathbf{y}) \nabla u(x, \mathbf{y})) &= g(x) & \text{in } U \subset \mathbb{R}^D, \\ u(x, \mathbf{y}) &= 0 & \text{on } \partial U, \end{aligned} \quad (127)$$

with $a : U \times \Gamma \rightarrow \mathbb{R}$ and $\Gamma := [0, 1]^d$.

We consider the case where $f(\mathbf{y})$ cannot be evaluated exactly because we do not have access to an analytic solution of the PDE. In this case, we solve the PDE with finite difference or finite element solver giving approximations of f : $f_n(\mathbf{y}) = Q(u_{\mathbf{y},n})$. The general methodology to apply multilevel least squares regression then proceeds as follows:

1. Solve the PDE with the coarsest resolution h_0 for a large number $|\Gamma_L|$ of randomly chosen values of the parameters $\mathbf{y}$.
2. Extend these results to the entire parameter domain by means of a weighted least squares approximation in the large polynomial subspace V_L .
3. Solve the PDE with a finer resolution h_1 and for the coarsest resolution h_0 for a smaller number $|\Gamma_{L-1}|$ of parameters $\mathbf{y}$.
4. Project the difference between coarse and fine resolution solutions onto the smaller polynomial subspace V_{L-1} .
5. Repeat for all levels l .
6. Sum all of the projections to obtain the final multi-level estimator for the function f .

Of critical importance is that the samples $\mathbf{y}$ used to compute f_l and f_{l-1} at a given level are the same. This guarantees that the sum (124) telescopes and our assumptions for the multi-level approximation are upheld.

11.5.2 Application to SDEs: Initial issues

We want to estimate:

$$(\bar{x}^0(t, s))^2 = \mathbb{E}[\bar{P}_1^T \bar{b}(t, \bar{X}(t)) \bar{b}(t, \bar{X}(t))^T \bar{P}_1 | \bar{P}_1 \cdot \bar{X}(t) = s, \bar{X}(0) = \bar{x}_0] \quad (128)$$

Where in the Black-Scholes model we have:

$$\bar{b}(t, X(t))_{ij} = X_i \Sigma_{ij} \quad (129)$$

Using the best mean square approximation we can relate finding an expectation value to an $\mathcal{L}^2$ regression problem:

$$\begin{aligned} \mathbb{E}[X|Y] &= g(Y) \\ g(Y) &= \arg \min_{h \in V} \mathbb{E}[(X - h(Y))^2], \quad V = \{h : \mathbb{E}[h(Y)^2] < \infty\} \end{aligned} \quad (130)$$

Single-Level:

Model coefficients in time and space domain:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \int_0^T \mathbb{E} \left[\left(\vec{P}^T b(t, \vec{X}(t)) b(t, \vec{X}(t))^T \vec{P} - h(t, \vec{P} \cdot \vec{X}(t)) \right)^2 | \vec{X}(0) = \vec{x}_0 \right] dt \quad (131)$$

And then apply a Monte Carlo estimator:

$$(\bar{b}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M} \sum_m \frac{1}{N} \sum_n \left(\vec{P}^T b(t_n, \vec{X}^m(t_n)) b(t_n, \vec{X}^m(t_n))^T \vec{P} - h(t_n, \vec{P} \cdot \vec{X}^m(t_n)) \right)^2 \quad (132)$$

where $\vec{X}^m(t)$ are independent realisations of $\vec{X}$. For ease of notation, define $g := \vec{P}_1^T \vec{b}(t, \vec{X}(t)) \vec{b}(t, \vec{X}(t))^T \vec{P}_1$.

$$\mathcal{S}_L(f) := \Pi_{V_L} g_0(\vec{X}^0) + \sum_{l=1}^L \Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) \quad (133)$$

where

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \| (g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\cdot) \| \quad (134)$$

where we have purposefully left the argument(s) of v blank. The key difference between the case of random PDEs and this SDE case is that the data is a function of **two** different parameters $\vec{X}^l$ and $\vec{X}^{l-1}$.

This gives us (at least) two options:

1. We **assume** that there exists a map $\theta_{\Delta t} : \vec{X}^l \rightarrow \vec{X}^{l-1}$ such that in the limit $\Delta t \rightarrow 0$, $\vec{X}^{l-1} = \vec{X}^l$. Then we can write:

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \| (g(\vec{X}^l) - g(\theta_{\Delta t}(\vec{X}^l))) - v(\vec{X}^l) \| \quad (135)$$

This map will be dependent on the particular path in question: $\theta_{\Delta t}(\vec{X}^l, \omega)$. The question is whether such a map exists and whether its properties allow for a telescoping sum to be written.

2. We allow that v is a function of two variables $v = v(\vec{X}^{l-1}, \vec{X}^l)$. Then we can write:

$$\Pi_{V_{L-l}} (g(\vec{X}^l) - g(\vec{X}^{l-1})) := \arg \min_{v \in V_{L-l}} \| (g(\vec{X}^l) - g(\vec{X}^{l-1})) - v(\vec{X}^{l-1}, \vec{X}^l) \| \quad (136)$$

Since the function to be estimate $\bar{b}^2(t, s)$ is a function of only one variable and assuming that X_l and X_{l-1} are close, we stipulate that $\bar{b}^2(t, s) = v(s, s)$. That is, we evaluate v along the diagonal. Important considerations will be i) determining an appropriate ansatz for v and ii) limiting the computational cost of doubling the dimension. We note that on the coarsest level, the assumption that X_l and X_{l-1} are close will be most strained since this is when the coarse and fine paths are furthest apart.

11.5.3 Application to SDEs: Option 1

We need:

1. Sequence of exponentially increasing polynomial subspaces $(V_l)_{l=0}^\infty$: We chose subspaces of (total degree) Legendre polynomials which are orthogonal on the interval $[-1, 1]$ with weight function $w = 1$.
2. Exponentially decreasing time steps $h_0, h_1, \dots, h_l, \dots$: We chose $h_l = h_0 \cdot 2^{-l}$.

We simulate M_l fine paths $X^f(t^l)$ and construct the coarse paths $X^c(t^{l-1})$ by summing the Brownian increments of the fine paths. We then have a choice:

1. Interpolate the coarse path on the missing fine time steps using a Brownian Bridge.

2. Regress the difference only on the coarse time intervals.

We assumed that the option 2 is sufficient. Further, we chose that $h = h(t_n, \vec{P} \cdot \vec{X}_f^m(t_n^c))$, so h was a function of the fine paths. For $l > 0$, the Multilevel Monte Carlo least squares estimator for the projected diffusion coefficient can then be written as follows:

$$(\bar{b}_l^{x_0})^2(\cdot, \cdot) - (\bar{b}_{l-1}^{x_0})^2(\cdot, \cdot) \approx \arg \min_{h \in V} \frac{1}{M_l} \sum_{m=1}^{M_l} \frac{1}{N_{l-1}} \sum_{n=1}^{N_{l-1}} \left([\vec{P}^T b(t_n, \vec{X}_l^m(t_n^{l-1})) b(t_n, \vec{X}_l^m(t_n^{l-1}))^T \vec{P} - \vec{P}^T b(t_n, \vec{X}_{l-1}^m(t_n^{l-1})) b(t_n, \vec{X}_{l-1}^m(t_n^{l-1}))^T \vec{P}] - h(t_n^{l-1}, \vec{P} \cdot \vec{X}_l^m(t_n^{l-1})) \right)^2 \quad (137)$$

11.5.4 Application to SDEs: Optimal Transport Map

In some applications, one seeks to construct a *measure-preserving transformation* from a one-dimensional continuous random variable X to another random variable Y , such that the push-forward of the law of X under the map coincides with the law of Y . More precisely, we look for a measurable map

$$T : \mathbb{R} \rightarrow \mathbb{R}, \quad \text{such that} \quad T_{\#} \mu_X = \mu_Y,$$

where μ_X and μ_Y denote the probability measures of X and Y , and $T_{\#} \mu_X$ denotes the push-forward measure. Among the infinitely many maps that may satisfy this property, *optimal transport* (OT) theory selects the one that minimizes a given transportation cost functional. A common choice in the quadratic case is

$$T^* = \arg \min_{T_{\#} \mu_X = \mu_Y} \mathbb{E} [|T(X) - X|^2],$$

which, under mild regularity assumptions, admits a unique monotone non-decreasing solution in the one-dimensional setting [9–11].

In practice, when only empirical samples $\{(x_i, y_i)\}_{i=1}^N$ are available, the OT map can be approximated via *discrete optimal transport algorithms*. These include the classical *Hungarian algorithm* for the exact discrete assignment problem and modern *entropic regularization* techniques [12], which lead to the *Sinkhorn algorithm*—a scalable method widely used in machine learning and data-driven applications. For the one-dimensional case, the OT map simplifies to a *quantile map*

$$T^*(x) = F_Y^{-1}(F_X(x)),$$

where F_X and F_Y are the cumulative distribution functions (CDFs) of X and Y , respectively. This formula provides an explicit and computationally trivial way to transport samples from X to the law of Y .